

Latency Robust Cooperative Perception using Asynchronous Feature Fusion

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Abstract

Recent advancements in cooperative perception have showcased substantial improvements compared to single-agent perception. Nonetheless, the inherent latency present in such systems can dramatically impair their effectiveness. In this paper, we propose a Latency Robust Cooperative Perception framework, named LRCP, to compensate for the effect of temporal asynchrony. The intuition of LRCP is to directly fuse asynchronous bird's-eye view (BEV) features instead of estimating aligned features. To achieve this, we first propose a novel flow prediction module that uses cached past BEV features to predict the flow with a non-discrete time delay at the BEV feature level. Then, the predicted flow is employed to guide the spatial sampling location of interests. Our approach substantially enhances the robustness of temporal asynchronous cooperative perception. Specifically, we achieved robust performance across a range of latencies up to 500 ms, with a performance degradation of only 1 percent point for AP@0.5 metric and 4 percent points for AP@0.7 metric at 500 ms on two public datasets (V2X-Sim and Dair-V2X). Code to reproduce our results is available at <https://github.com/JesseWong333/LRCP>.

1. Introduction

Cooperative perception [2, 9] involves the integration of sensory data from multiple agents to generate a comprehensive and accurate representation of the surrounding environment. Unlike single-agent perception systems, cooperative perception harnesses the power of collective intelligence to enhance perception performance and address challenges like occlusions, sensor noise, and ambiguities, which can be widely applied to various domains, including autonomous driving and robotics sensing. Recent studies [4, 12, 20, 22, 26, 32] have shown the advantage of cooperative perception on 3D object detection and semantic segmentation tasks over single-agent perception systems, and also achieved a trade-off between communication bandwidth and perception performance [11].

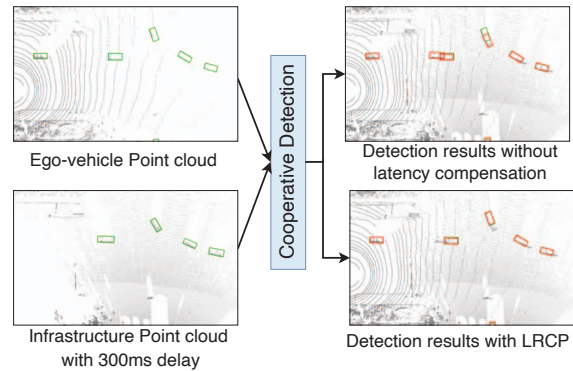


Figure 1. An example of asynchronous cooperative perception. Without latency compensation, the 3D detection may suffer from duplicate or inaccurate detections. LRCP can improve the detection performance when latency exists. Red boxes: detected, Green boxes: ground-truth

Despite cooperative perception offers numerous benefits over single agent-based perception, the inherent latency in such systems can significantly hinder their effectiveness, as illustrated in Fig. 1. The latency can be attributed to a number of factors, such as data processing and communication delays, which are inevitable in a realistic environment. Recent methods primarily rely on intermediate fusion techniques to aggregate bird's-eye-view (BEV) features from different agents, while latency issues result in the misalignment of BEV features. Although efforts have been made to mitigate the delay effects [16, 27, 32, 37], a substantial gap still remains when compared to the ideal scenario with no latency. We identified two key limitations in existing methods that may hinder compensation performance: 1) the absence of explicit motion representation between frames and 2) the potential accumulation of errors. Encoding delay information, as proposed in [32], does not directly capture motion dynamics, which may limit compensation effectiveness. While estimating aligned BEV features [16] offers a potential solution, it does not explicitly model intrinsic motion and may introduce additional noise to the features. Approaches that predict flow using region of interest (ROI) proposals from a single agent and then generate aligned BEV features through warping, such as [27], are suscep-

tible to error accumulation due to the sequential nature of the process.

In this paper, we propose a Latency Robust Cooperative Perception framework (LRCP) to mitigate the challenges posed by temporal asynchrony in cooperative perception, which aims to address the two key limitations of existing methods. Our core idea is to directly fuse asynchronous BEV feature maps instead of estimating temporally aligned feature maps. This is implemented through deformable operators [5, 29]. Deformable operators predefine a set of *reference points* and predict offsets to augment the spatial sampling locations. The offsets are learned based on the target tasks (e.g. classification task, detection task) in an end-to-end manner, with no additional supervision required. To achieve asynchronous feature fusion, we adjusted the locations of predefined reference points, a procedure we termed *reference points calibration*. We further propose a novel flow prediction module that uses cached past BEV features to predict the flow with a non-discrete time delay at the BEV feature level. This estimated flow is then used to calibrate the reference points. The proposed flow prediction module leverages deformable attention to aggregate multiple cached BEV features and encodes latency information into flow queries, enabling non-discrete flow prediction. By directly generating flow from cached BEV features and bypassing the additional feature generation steps, LRCP potentially reduces the accumulation of errors, leading to more robust and accurate perception performance.

We have done extensive experiments on V2X-Sim dataset [19] and Dair-V2X [35] to evaluate the effectiveness of LRCP. The results verify the robustness of our framework under varying latency and show substantial improvements over existing methods. In summary, the contributions of this work include:

- We propose a new Latency Robust Cooperative Perception (LRCP) framework to address the challenge of temporal asynchrony in cooperative perception, which captures the motion intrinsic and aims to reduce the accumulative errors.
- We propose a novel flow prediction module that uses cached past BEV features to predict a flow with non-discrete time delay.
- Our evaluations demonstrate robust performance across a range of latencies up to 500 ms on the V2X-Sim and Dair-V2X datasets. Compared to the ideal scenario with no latency, LRCP exhibits a minimal performance degradation, maintaining an AP@0.5 score within 1 percent point and an AP@0.7 score within 4 percent points (p.p.) at 500 ms latency.

2. Related works

Our work addresses the challenges of temporal asynchrony in cooperative perception. This approach leverages scene flow prediction to compensate for the latency issue, which is related to several areas, including scene flow prediction itself, cooperative perception in general, and the latency issue specifically.

2.1. Scene flow prediction

Scene flow describes the 3D motion of objects in a scene over time. When this motion is recovered from a 2D plane, it can also be referred to as 2.5D scene flow [38]. Extensive research has been done in the field of 3D scene flow [8, 17, 39], and also in the field of 2.5D scene flow [10, 28]. Notably, scene flow estimation can also be approached using self-supervised methods [17]. Existing methods usually predict the scene flow of two given frames or predict the flow in a fixed time interval. In contrast, our proposed flow module encodes the time information and can predict the flow with non-discrete time with the same head. Furthermore, our approach simplifies the process by predicting flow at the BEV feature level, enhancing efficiency and flexibility.

2.2. Cooperative perception

Cooperative perception allows multi-agents to share perceived information through the communication network, significantly improving the perception capabilities over single-agent perception. Both simulated datasets [19, 33] and real-world datasets [31, 35, 42] have been developed to support algorithm development. Researchers have made significant progress in addressing challenges inherent to cooperative perception systems, such as communication bandwidth limitations [11, 13, 20], localization errors [22, 24], latency issues [16, 27, 36], and the presence of heterogeneous devices [18, 30, 32]. Additionally, some research focuses on camera-only methods [12] for 3D object detection and semantic occupancy prediction [23]. Since low-latency perception is essential for the safety of autonomous vehicles, we aim to tackle the latency challenge in cooperative perception.

2.3. Cooperative perception with latency

Several studies have addressed the challenge of latency in cooperative perception. V2X-ViT [32] incorporates delay-aware positional encoding to consider time delay information, while SyncNet [16] employs an LSTM-based model to predict aligned feature maps from cached features. However, both approaches lack the ability to explicitly represent motion between frames. Alternative approaches focus on generating flow to recover aligned feature maps. Examples of this include CoBevFlow [27], which predicts trajectories based on detected motion-related attributes, and

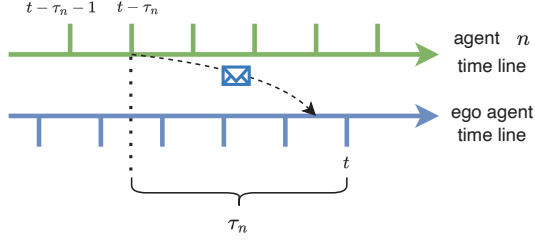


Figure 2. Illustration of the time latency τ_n between ego agent and agent n

How2comm [34], which densely predicts the flow map for a fixed time interval. While existing methods enhance performance across varying latency levels, a significant gap still exists compared to the ideal scenario. In LRCP, we directly fuse asynchronous BEV by calibrating the reference points in deformable attention. This eliminates the need for aligned feature generation, as used in existing works, leading to significantly increased robustness across a range of latencies.

3. Methodology

Section 3.1 introduces cooperative perception tasks and the concept of temporal asynchrony in cooperative perception. Subsequently, Section 3.2 describes the overall architecture of the proposed LRCP framework, detailing the calibration of reference points in deformable attention and the flow prediction module. Finally, Section 3.3 elaborates on the training scheme employed.

3.1. Cooperative perception and temporal asynchrony

Cooperative perception is a technique enabling multiple agents to collectively perceive a scene by exchanging perceived information through wireless communication, e.g. V2X communication [1]. Due to the inevitable processing and communication delays in real-world scenarios, fusing features from differing timestamps often leads to a misalignment issue. Consider a scenario with N agents in the same scene, each agent $n \in \{1 \dots N\}$ senses the environment using its own sensors, and processes the data captured at a specific timestamp t_n to obtain a feature map $\mathbf{F}_n^{t_n}$. Subsequently, these processed features, along with the timestamp and the agent's pose, are broadcasted through a communication network. The feature extraction method remains consistent with the single-agent approach [15, 40]. Features received from other agents are then projected to the ego agent's coordinate system, denoted as $\mathbf{F}'_n^{t_n}$. In our context, we define τ_n to represent the latency in cooperative perception, as illustrated in Fig. 2. Note that the time interval between two consecutive frames for any agent is constant as the agent's sensors operate at a certain sampling rate, considering no frame lost during the communication.

And the latency τ_n is non-discrete as the sampling timestamps of different agents are not aligned. The latency τ_n in our experiments can vary from a minimal delay of less than 100 ms to a maximum of 500 ms. A robust latency compensation method should maintain performance during low latency while also improving performance under high latency. Finally, the ego agent aggregates all features to generate a fused feature map $\hat{\mathbf{F}}_{ego}^t$ at the ego's coordinate system. The fused feature map is then fed into a detection head $f_{head}(\cdot)$ to get the final result $\mathbf{Y}_{ego}^t$. The cooperative perception tasks with latency can be formulated as follows:

$$\mathbf{F}_n^{t_n} = f_{encoder}(\mathbf{X}_n^{t_n}), n \in \{1 \dots N\}, \quad (1)$$

$$\mathbf{F}'_n^{t_n} = project(\mathbf{F}_n^{t_n}), \quad (2)$$

$$\hat{\mathbf{F}}_{ego}^t = f_{agg}(\mathbf{F}_{ego}^t, \{\mathbf{F}'_{n,n \neq ego}^{t-\tau_n}, \dots, t-\tau_{n-k-1}, \tau_n\}), \quad (3)$$

$$\mathbf{Y}_{ego}^t = f_{head}(\hat{\mathbf{F}}_{ego}^t). \quad (4)$$

Here, we use the minus operation in time to denote a previous frame, k is the number of cached frames, and $f_{agg}(\cdot)$ is the aggregation operator.

3.2. Latency Robust Cooperative Perception framework

The direct fusion of features from different timestamps can markedly compromise performance, particularly in scenarios involving moving objects. In this section, we present a novel latency compensation framework for cooperative perception, which can effectively fuse asynchronous features and maintain strong robustness to various latency levels.

Overall Architecture. As illustrated in Fig. 3, our framework follows an encoder-decoder architecture [25, 41], which consists of a feature encoder, an asynchronous feature fusion decoder with deformable cross/self-attention layers, and a task-specific head. In addition to that, we integrate a flow prediction module aimed at calibrating the reference points utilized in deformable cross-attention layers. For each agent n , the feature encoder processes the raw data at timestamp t_n and produces BEV features $\mathbf{F}_n^{t_n} \in \mathbb{R}^{H \times W \times C}$. Similar to [21], we first initialize a set of grid-shaped learnable parameters $Q \in \mathbb{R}^{bev.h \times bev.w \times bev.c}$ as BEV queries, where $bev.h$, $bev.w$ are the spatial dimensions of the BEV plane, each grid $q_{p=(x,y)} \in \mathbb{R}^{1 \times bev.c}$ within the BEV queries representing the feature within a spatial grid at location $p = (x, y)$. Additionally, positional embeddings are added to BEV queries Q before being sent to the fusion decoder. Our fusion process in step 2 is based

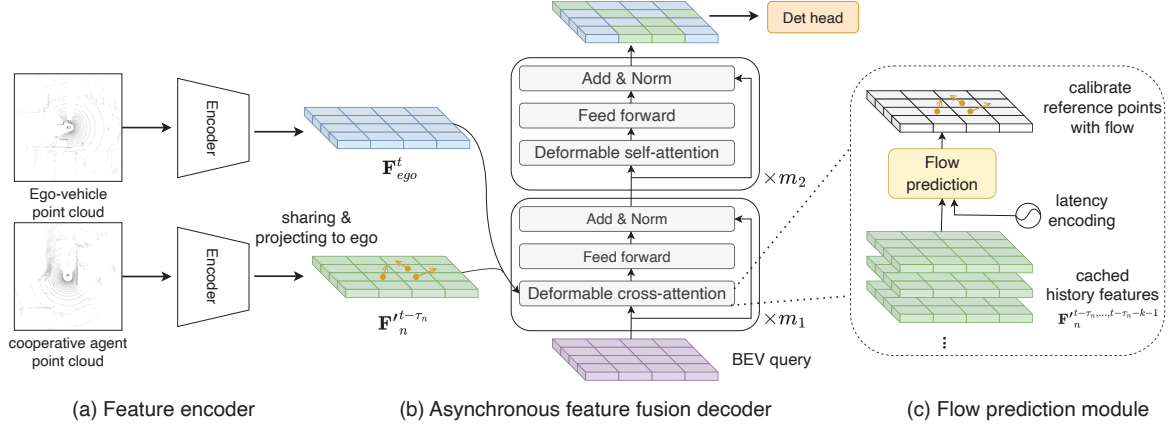


Figure 3. **Overall architecture.** (a) The feature encoder transforms the raw input into BEV features. (b) We apply deformable attention to aggregate multiple BEV features. Reference point calibration is only applied to non-ego agents, while reference points of the ego agent remain unchanged. (c) The reference points calibration module is fed into k cached features $\mathbf{F}_n^{t-\tau_n, \dots, t-\tau_n-k-1}$ and the latency τ_n to predict a flow, which is then used to calibrate the reference points.

on deformable attention, which can be formulated as:

$$\text{DeformAttn}(q, \{p_n\}_{n=1}^N, \{\mathbf{F}_n\}_{n=1}^N) = \sum_{i=1}^{N_{\text{head}}} \mathcal{W}_i \left[\sum_{n=1}^N \sum_{j=1}^{N_{\text{key}}} \mathcal{A}_{ijn} \cdot \mathcal{W}'_i \mathbf{F}_n(p_n + \Delta p_{nij}) \right], \quad (5)$$

where N_{head} is the total number of attention heads, N_{key} is the total sampled key number for each head. $\mathcal{W}_i$ and $\mathcal{W}'_i$ are the learnable weights. $\mathcal{A}_{ijn} \in [0, 1]$ is the predicted attention weight, and is normalized by $\sum_{n=1}^N \sum_{j=1}^{N_{\text{key}}} \mathcal{A}_{ijn} = 1$. $\Delta p_{nij} \in \mathbb{R}^2$ are the predicted offsets to the reference point p_n . For simplicity, we omit the superscript of F_n .

We use different reference points p_n for different agents to guide the aggregation process. These reference points p_n are calibrated by the predicted flow to ensure accurate alignment. After multiple deformable cross-attention layers, we then use deformable self-attention layers to progressively refine the features and get the fused BEV feature. Finally, the resulting fused BEV features are directly fed into a 3D detection head to generate the final detection results.

Calibrate the reference points. For the received feature $\mathbf{F}_{n, n \neq \text{ego}}^{t-\tau_n}$ from a non-ego agent, we calibrate the reference points p_n to align feature maps from different timestamps. The calibration process is shown in Fig. 4. We use a flow prediction module to predict a 2D flow map $\text{flow}_n \in \mathbb{R}^{b_{\text{ev.h}} \times b_{\text{ev.w}} \times 2}$ from timestamp t to timestamp $t - \tau_n$. The calibrated reference points are obtained through:

$$\text{flow}_n = f_{\text{calibrate}}(\mathbf{F}_n^{t-\tau_n, \dots, t-\tau_n-k-1}, \tau_n) \quad (6)$$

$$p_n = p'_n + \text{flow}_n \quad (7)$$

where p'_n is the default reference point, and the value is the location of the query. This calibration process enables

the deformable fusion layers to directly extract information from the correct spatial locations within the received features $\mathbf{F}_n^{t-\tau_n}$, facilitating the alignment without the need for explicit feature generation. By integrating the reference point calibration process into the pipeline, the system gains the ability to effectively synchronize feature maps from different timestamps.

Flow prediction module. As described above, our framework relies on k BEV feature maps $\mathbf{F}_n^{t-\tau_n, \dots, t-\tau_n-k-1}$ and the latency τ_n to predict a flow from timestamp t to $t - \tau_n$, which is achieved by the proposed flow prediction module. The module's design addresses two key challenges: 1) Non-discrete temporal prediction: Unlike MotionNet [28] which employs multiple regression heads for discrete timestamps, this module necessitates predicting flow for any given non-discrete time interval, accounting for the varying nature of latency. 2) Efficient temporal feature aggregation: While 3D convolutions [3, 28] represent a common approach for capturing temporal relationships, their computational cost can be significant. To tackle the identified challenges, we propose a novel flow prediction module that extracts multiple temporal features to enable non-discrete flow prediction, as illustrated in Fig. 5. For each feature map $\mathbf{F}_n^{t-\tau_n, \dots, t-\tau_n-k-1}$, a learned positional encoding is incorporated to encode the frame order. Similar to BEV queries, we initialize a group of learnable parameters $Q_{\text{flow}} \in \mathbb{R}^{b_{\text{ev.h}} \times b_{\text{ev.w}} \times b_{\text{ev.c}}}$ as flow queries, a positional embedding is also added in the queries. Additionally, we leverage latency encoding to encode the latency τ_n into the flow queries, facilitating the prediction of flows for non-discrete timestamps. For the latency encoding, we use

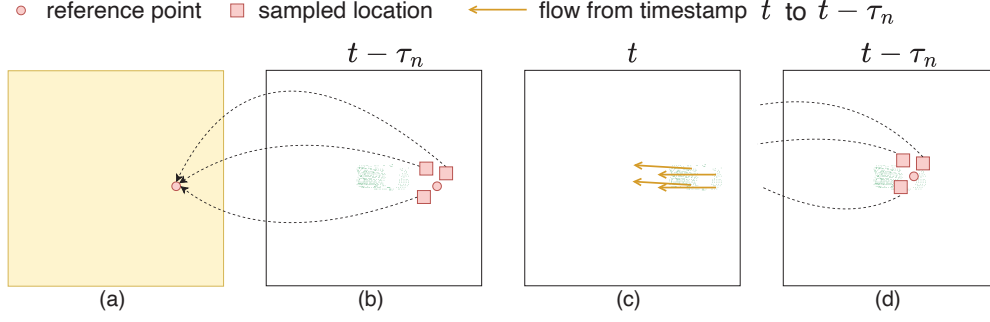


Figure 4. **Illustration of the reference point calibration process** (a) The flow query map. (b) Without calibration, the reference point falls into the wrong position. (c) At timestamp t , we estimate a flow from timestamp t to $t - \tau_n$. (d) The reference point is calibrated by the estimated flow and is now positioned correctly.

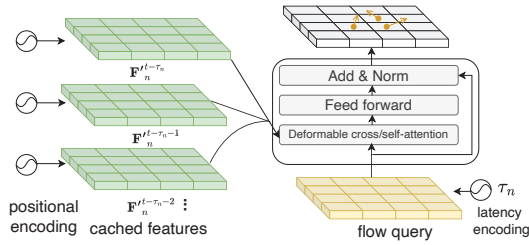


Figure 5. Illustration of flow prediction module
the same sinusoidal encoding as in [25],

$$\begin{aligned} LE(\tau_n, 2c) &= \sin(\tau_n / 10000^{\frac{2c}{bev.c}}) \\ LE(\tau_n, 2c + 1) &= \cos(\tau_n / 10000^{\frac{2c}{bev.c}}) \end{aligned} \quad (8)$$

where c indexes the channels $bev.c$ of flow queries. To achieve efficient multi-temporal feature aggregation, this study adopts deformable cross/self attention as an alternative to 3D convolutions. This strategy reduces the computational complexity from $O(kHWd^3C^2)$ to $O(N_qC^2 + kHWC^2)$, where d is the dimension of the 3D convolution kernel size. This architectural refinement ensures effective temporal feature aggregation while minimizing computational overhead, thus enhancing the overall efficiency and performance of the proposed framework.

3.3. Training scheme

Our proposed asynchronous feature fusion method employs a two-stage training strategy to ensure effective learning and model convergence. **Stage 1:** The initial phase focuses on training the feature encoder, decoder, and detection head specifically for the detection task. During this stage, the flow module remains untrained, and the calibration of reference points within the deformable attention mechanism utilizes the ground truth flow. **Stage 2:** We maintain the previously trained components, feature encoder, decoder, and detection head unchanged, while the flow module is trained under the supervision of the ground truth flow. Adopting this structured training approach allows for faster model

convergence, as the initial focus on detection accuracy sets a strong baseline for subsequent flow module training.

4. Experiments

In this section, we detail the implementation and experimental setup of the proposed asynchronous cooperative perception framework. Subsequently, we evaluate the 3D detection performance under varying latencies on two benchmark datasets to validate the effectiveness of our compensation approach. Additionally, we conduct ablation studies to show the effectiveness of the designs. Finally, we present visualization results to illustrate the efficacy of our methods qualitatively.

4.1. Datasets

V2X-sim [19]. V2X-Sim is built with the co-simulation of SUMO [14] and CARLA [6]. We follow the updated V2X-Sim datasets, which include 80 scenes in the training set, 10 scenes in the validation set, and 10 scenes in the test set. Each scene has a 20-second duration and involves $N = 2, 3, 4, 5$ vehicles as intelligent agents for information exchange. Following the setting of [22], we only consider vehicle-to-vehicle communication. The stream data is recorded at a frequency of 5 Hz, yielding a total of 10,000 frames. To simulate cooperative perception with temporal asynchrony, we load data from previous time stamps. Note that, as there are no previous frames at the start of a scene, the number of frames used in training and evaluating is slightly reduced from the total number of frames.

Since the original V2X-Sim is focused on detection and segmentation tasks, motion labels for objects are not provided. To provide supervision for our flow module, we generate 2D flow maps between two frames as follows: We first project the grounding truth bounding boxes to the ego agent's coordinated system, then use the unique object ID to track one object across the two frames. Among the two bounding boxes of one object, we calculate the relative translation $\Delta x, \Delta y$, and yaw angle $\Delta \theta$ to obtain a trans-

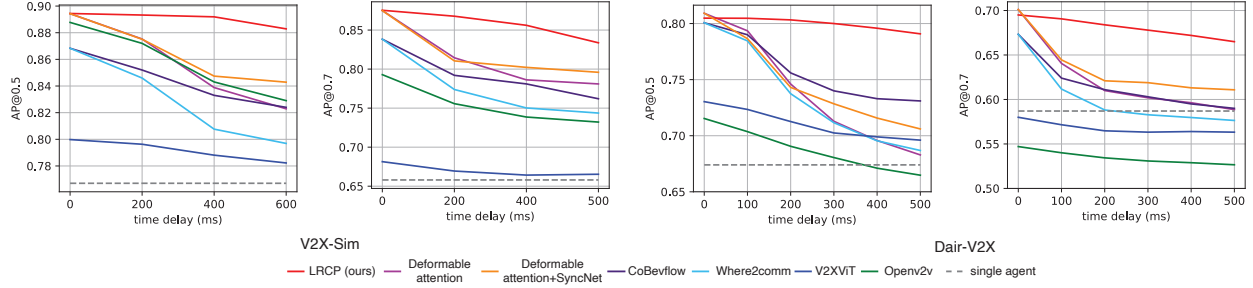


Figure 6. Comparison of 3D detection performance on the V2X-Sim and Dair-V2X datasets under latencies ranging from 0 to 500/600 ms. Detailed tabular results can be found in the appendices.

formation matrix $T = \begin{pmatrix} \cos(\Delta\theta) & -\sin(\Delta\theta) & \Delta x \\ \sin(\Delta\theta) & \cos(\Delta\theta) & \Delta y \end{pmatrix}$. The transformation T is then applied to all points within the source bounding boxes. By subtracting the original location, we can get the flow within the bounding boxes. Points not enclosed within any bounding boxes are considered stationary, and their corresponding flow values are set to zero.

DAIR-V2X [35]. DAIR-V2X is a real-world cooperative perception dataset. Each frame within DAIR-V2X encompasses one ego vehicle agent and one infrastructure agent. In total, DAIR-V2X provides a collection of 6,617 frames for cooperative perception, each captured from real-life situations at a rate of 10 Hz with 3D annotations. We adopted the annotations as in [22], which complemented the missing label to cover 360-degree detection. Since DAIR-V2X doesn't provide the flow annotations, we therefore adopted an annotation approach similar to the V2X-sim dataset except for the following steps: We used a greedy method to track individual objects across multiple frames due to the lack of a unique track ID. Then we used a gradient descent method to estimate the transformation between the two bounding boxes because we noticed that the size and shape of the two bounding boxes were not consistent. Simply using the center points and the yaw angle may yield an inaccurate transformation.

4.2. Implementation details

For the V2X-sim dataset, a random agent is selected as the ego agent during training, while we evaluate a fixed ego agent during testing. For the DAIR-V2X dataset, the vehicle side is always selected as an ego agent. We use PointPillar [15] with a ResNet backbone as the feature encoder, and the voxel resolution is 0.4 m. The detection range is $\pm 32 \text{ m} \times \pm 32 \text{ m}$ for V2X-sim dataset and $\pm 140 \text{ m} \times \pm 40 \text{ m}$ for the DAIR-V2X dataset. We stacked 4 deformable cross-attention layers and 2 deformable self-attention layers as the asynchronous feature fusion decoder. Each layer has 8 heads and 8 sampling points. For the flow prediction module, we use 3 cross-attention layers and 2 self-attention layers with 8 heads and 8 sampling points by default. We

adopted the multi-scale feature strategy to aggregate intermediate features at multiple spatial resolutions. All collaborators in our experiments follow the 1-round communication setting [22, 33], in which each agent shares the pose and feature map simultaneously. We used the same anchor-based detector head and losses for 3D object detection task as in [11, 22, 32, 33], with an initial learning rate of 0.002 to train the detection model at stage 1. For each agent, we also added a separate detection head to supervise the learning in the agent. We use the weighted SmoothL1Loss with a learning rate of $5e^{-4}$ to train the flow model at stage 2. Background and foreground weighting values were set at 0.005 and 1, respectively. Evaluation is based on the Average Precision (AP) metric [7] at the specified Intersection-over-Union (IoU) thresholds of 0.5 and 0.7.

4.3. Evaluation results

Fig. 6 shows the 3D detection performance on V2X-Sim and Dair-V2X datasets, in comparison to benchmark models including openv2v [33], where2comm [11], Cobeflow [27], SyncNet [16] and V2XViT [32]. *Deformable attention* is our implementation of using deformable attention to aggregate multiple agent features but without reference point calibration. For the V2X-sim dataset, we consider latency up to 600 ms instead of 500 ms, since the data is recorded at 5 Hz and we therefore have 200 ms between samples. We see that deformable attention is a strong baseline, which achieves state-of-the-art performance on both of the datasets when the latency is zero. However, its effectiveness diminishes under latency, with a notable decline by 7.1 p.p. and 8.6 p.p. for AP@0.5 and AP@0.7 metrics respectively on the V2X-Sim dataset at 600 p.p. latency, and 12.6 p.p. and 11.3 p.p. on the Dair-V2X dataset at a 500 ms latency. In contrast, the proposed LRCP substantially improves the robustness of temporal asynchronous cooperative perception under various latencies. When encountering latencies below 200 ms, the performance reduction is negligible. When at the extreme case (500/600 ms latency), the performance degradation is restricted to only

Table 1. More comparison of performance degradation at 400 ms latency for AP@70 metric with existing official implementations.

Method	Conference	Datasets	Ave. degradation (p.p.)
V2X-ViT [32]	ECCV 2022	V2XSet	11.9
SynNet [16]	ECCV 2022	V2X-Sim	4.5
CoBeVflow [27]	NeurIPS 2023	IRV2V, DAIR-V2X	9.2
How2comm [34]	NeurIPS 2023	V2XSet, OPV2V, DAIR-V2X	14.1
LRCP (ours)	-	V2X-Sim, DAIR-V2X	2.3

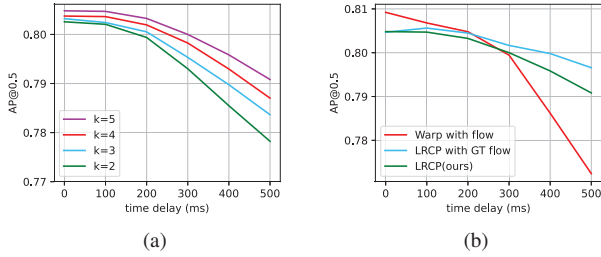


Figure 7. (a) Ablation of past frames k . b) Comparison of LRCP, LRCP with ground truth flow and warping method.

Table 2. Effect of the number of self/cross attention layers in flow prediction module

layers	AP@0.5(500 ms)	AP@0.7(500 ms)
3 cross	78.6	65.9
2 cross+1 self	78.9	66.6
5 cross	78.9	66.3
3 cross+2 self	79.1	66.5

1 p.p. for AP@0.5 and less than 4 p.p. for AP@0.7 on both datasets. Compared with the benchmark models, LRCP can even surpass some methods that are already well synchronized, highlighting its superior capability in handling temporal asynchronous communication.

Due to the variation in experimental setups, fair comparisons between methods as described in recent literature are difficult to achieve. Still, to provide a more comprehensive picture, we incorporate more comparison of performance degradation results with existing official implementations, as shown in Tab. 1. LRCP outperforms existing methods by exhibiting minimal performance degradation.

Fig. 7a illustrates the impact of varying the number of cached past frames k . As observed, employing a larger k consistently yields significantly better results compared to using fewer frames. To achieve a balance between computational overhead and performance, we empirically set $k = 5$ as the default value. Tab. 2 presents an ablation study on the number of self-attention and cross-attention layers within the flow prediction module. When maintaining a constant overall number of layers, integrating both self and cross-attention layers proves more effective than solely relying on self-attention layers. Moreover, incrementally

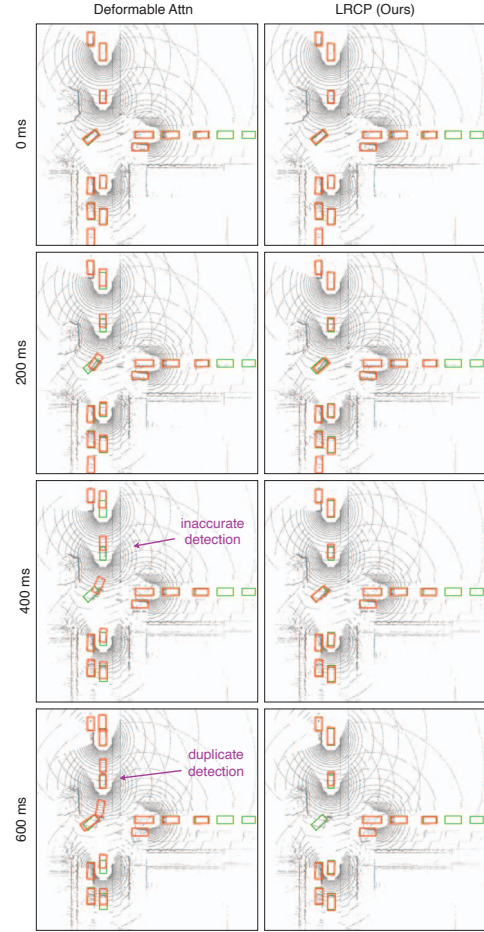


Figure 8. Visualization of the detection results on v2x-sim datasets. We show the detection results of deformable attention and LRCP when the latencies are 0 ms, 200 ms, 400 ms, and 600 ms. Red boxes: detected, Green boxes: ground-truth

stacking more layers offers marginal performance improvements. Further comparisons of the proposed method under various settings are presented in Fig. 7b.

In LRCP with ground truth flow, the ground truth flow is used for reference point calibration. In warping with flow, a warping method, as in [27], with the predicted flow from our flow prediction module is used to generate an aligned feature map. We see that: 1) employing the predicted truth flow instead of ground truth flow in LRCP results in a slight degradation in performance (0.5 p.p.), which also validates the effectiveness of the flow prediction module. 2) Simply relying on warping yields a performance drop of 2.5 p.p. compared to LRCP, potentially attributable to noise introduced by the warping method. More ablation studies can be found in the Appendix.

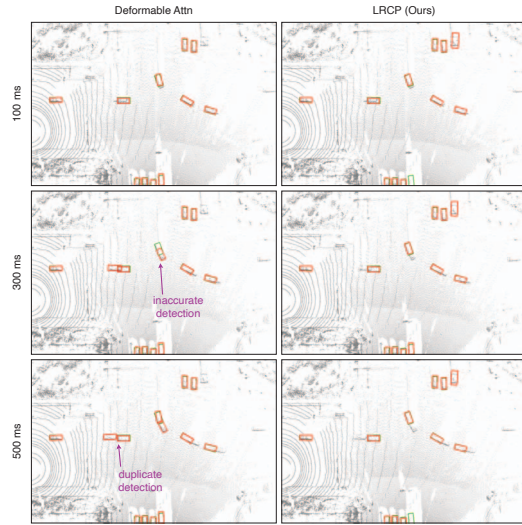


Figure 9. Visualization of the detection results on DAIR-V2X datasets. We show the detection results of deformable attention and LRCP when the latencies are 100 ms, 300 ms, and 500 ms. **Red boxes:** detected, **Green boxes:** ground-truth

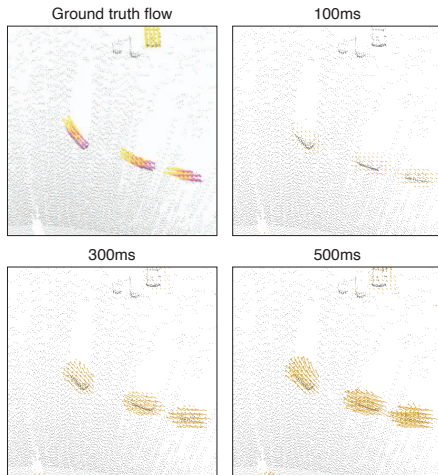


Figure 10. Visualization of the flow map under 100 ms, 300 ms and 500 ms latency. The ground truth flow for various timestamps is illustrated using a gradient color scheme, while the predicted flow is indicated by the arrow lines.

4.4. Qualitative evaluations

Fig. 8 and Fig. 9 visualize the detection results under various latencies on the V2X-Sim dataset and Dair-V2X dataset, both with and without compensation. We see that both methods achieve accurate object detection performance at low latency. As the latency increases, deformable attention begins to exhibit erroneous detections. These errors manifest in two main types: inaccurate detection, typically occurring under moderate latency, where the predicted boxes from the past frame suppress those of the current

frame; and duplicate detection, typically occurring when the latency is high, where the same object is detected in both the previous and current frames. The visualization results show that LRCP can effectively compensate for both types of errors. Fig. 10 visualize the flow prediction results. When encoding different latencies, the flow prediction module can generate a non-discrete flow. We observe that our flow prediction module accurately predicts the direction and magnitude of the flow. We also observed minor inconsistencies within the flow predictions, such as jitters where backgrounds adjacent to objects exhibit unintended motion. Developing strategies to refine flow predictions will be an essential focus of our future work.

5. Limitations and future works

The proposed framework aims to enhance the latency robustness of cooperative perception systems through the integration of V2X communication. However, several challenges persist in real-world scenarios. For instance, V2X communication faces limited bandwidth constraints, while pose estimation for moving agents often introduces noise. Exploring solutions to address these issues within our framework can be further explored. Moreover, a disparity remains compared to utilizing ground truth flow, necessitating further exploration into predicting non-discrete timestamp flows. Additionally, emerging concerns regarding privacy and adversarial robustness during data sharing require greater attention, as they have not been thoroughly addressed.

6. Conclusion

In conclusion, this work presents LRCP, a novel framework designed to address the challenges posed by temporal asynchrony in cooperative perception systems. LRCP achieves this by integrating two key components: a flow prediction module to capture inter-frame motion and a deformable attention mechanism that avoids the generation of temporally aligned features, thereby minimizing cumulative errors. Through extensive evaluations on benchmark datasets, LRCP demonstrates significant improvements in robustness against varying latency conditions. Even under severe latency, the framework exhibits only marginal performance degradation, highlighting its efficacy in mitigating the detrimental effects of temporal asynchrony. This paves the way for the development of more reliable cooperative perception systems for real-world applications.

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